

HARDHIK Y

Bengaluru , India · hardhiky9@gmail.com · 9620406339 · <https://hardhik.vercel.app/>

EDUCATION

PES University

Btech Computer science *GPA: 7.54*

Bengaluru , India
Aug 2022 - May 2026

WORK EXPERIENCE

Ellipsonic

Software Engineer Intern

Bengaluru India
June 2024 - August 2024

- Engineered a custom OCR pipeline (OpenCV, PaddleOCR, ML) to extract data from pipe diagrams, exposed via REST API (FastAPI + PostgreSQL).
- Automated QA checks and async processing with Celery + Redis, reducing manual effort by 40%.
- Collaborated with client teams to integrate solution into existing workflows.

CIE – PES University

Machine learning intern

Bengaluru India
January 2025 - May 2025

- Developed a real-time object detection system (OpenCV + PyTorch/TensorFlow) deployed as a scalable API service with Docker.
- Optimized inference with CUDA and batching, cutting latency by >60% for live CCTV streams.
- Built automated data pipelines and monitoring dashboards for retraining and production reliability.

Springer Capital

Software Engineer Intern (CRM)

Chicago, Illinois
May 2025 - August 2025

- Built multi-tenant CRM backend with Django REST Framework, PostgreSQL, and API versioning.
- Implemented async jobs (Celery + Redis) for reporting and bulk operations; reduced runtime from hours to minutes.
- Developed Vue.js SPA with WebSockets (Django Channels) for real-time notifications and activity logs.
- Hardened system with OAuth2/JWT, input validation, and monitoring (Sentry, Prometheus).
- Automated deployments via Docker + GitHub Actions CI/CD, improving release velocity by 30%.

SKILLS

Languages:	C/C++, Java, Python, Rust, JavaScript, TypeScript, SQL (PostgreSQL), OCaml, Go
Framework:	Django REST , Spring , React, Next.js, Vue.js, Svelte, Node.js, Express, FastAPI
Libraries:	Pandas, NumPy, Matplotlib, OpenCV, Zod, Framer Motion, shadcn/ui, Prisma, tRPC, Vite
AI/ML tools:	PyTorch, TensorFlow, PaddleOCR, Transformers, LangChain, OpenCV
Cloud & Infra:	AWS (S3, Lambda, ECS), Docker, Kubernetes, Terraform, CI/CD (GitHub Actions)
Systems & Tools:	QEMU, GDB, Make, CMake, Git, Prometheus, Grafana, Sentry

PROJECTS

AI Interview Bot (Final Year Project) *Python, React, Node.js, TensorFlow, PostgreSQL*

(<https://github.com/Hardhiky/capstone>)

Developed an AI-powered voice bot to assist candidates in technical interviews, featuring an adaptive questioning system that adjusts difficulty based on user responses. Implemented resume parsing to generate interview questions tailored to user skills. Integrated real-time Speech-to-Text (STT) and Text-to-Speech (TTS) for seamless interaction. Built an AI model leveraging NLP and semantic similarity techniques to evaluate answers. Designed a user-friendly frontend with React and a robust backend API using Node.js and Express.

Drawfish *Python, ML, React js , Node js, Git*

(<https://github.com/Hardhiky/drawfish>)

Developed a custom Stockfish model integrated with a proprietary ML model to enforce drawish positions in openings, middle game, and endgame. Engineered the AI to ensure every game results in a draw, regardless of the player's moves. Utilized advanced chess heuristics and reinforcement learning to refine decision-making. Optimized performance for real-time gameplay and strategic balance.

RIX *Rust , Assembly , Qemu , GDB, Make, CMake*

(<https://github.com/Hardhiky/rix>)

RIX is an experimental operating system for the Intel IA-32 architecture (x86). It's written completely from scratch in Rust and doesn't use any external dependencies.

Cloud-Native Document Search Platform *Next.js 14 (App Router), Tailwind CSS, shadcn/ui, Prisma, PostgreSQL + pgvector, Clerk, Tiptap, Y.js (CRDT), OpenAI/Mistral, LangChain, IndexedDB, Service Workers (PWA), Vercel, Turborepo, GitHub Actions, UploadThing, and Cloudflare R2.*

Designed an event-driven ingestion pipeline (S3 → Lambda → pgvector + PostgreSQL) for OCR, embeddings, and semantic search. Deployed autoscaling APIs on GCP Cloud Run/Kubernetes; infra managed via Terraform. Implemented RBAC/IAM policies, observability with Prometheus + Grafana, and CI/CD pipelines.

AI-Assisted Code Review Tool *Svelte + OCaml*

Built a static analysis engine in OCaml (AST parsing, type inference, rule-based checks), exposed via REST + WebSocket API. Integrated with GitHub Actions for PR gating and code quality enforcement. Frontend: Svelte + Monaco Editor with inline suggestions, diff highlighting, and live analysis.

AI Knowledge Base & Chat Platform *Next.js, LangChain, FastAPI, PostgreSQL, Redis, OpenAI/Mistral, Docker, Kubernetes*

Designed and deployed a cloud-native AI-powered knowledge management platform where users can upload PDFs, docs, and web content, which are then indexed into pgvector (PostgreSQL) for semantic search and real-time retrieval. Built a scalable FastAPI backend serving both REST and WebSocket endpoints for chat sessions; integrated LangChain for retrieval-augmented generation (RAG) pipelines with OpenAI/Mistral LLMs. Frontend: Next.js 14 + Tailwind + shadcn with a collaborative chat interface supporting multi-user sessions and streaming AI responses. Optimized performance using Redis caching, async task queues (Celery/RQ), and background workers for document preprocessing. Deployed on Kubernetes (GKE) with autoscaling; set up CI/CD via GitHub Actions, monitoring with Prometheus + Grafana, and error tracking with Sentry. Industry Keywords: semantic search, vector embeddings, RAG, multi-tenant SaaS, observability, real-time chat, Kubernetes autoscaling, Redis caching, LangChain, LLMOps.

ACHIEVEMENTS

KODIKON 3.0 (A national level hackaton)

achieved a Top 10 position in a national-level hackathon, demonstrating strong problem-solving, teamwork, and technical skills in a competitive environment.

Nov 2024

Open-source projects

Published multiple open-source projects (RIX OS, AI Interview Bot, Drawfish engine).